

Notes: Graham (JF, 2000)

How Big Are the Tax Benefits of Debt?

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1 Big picture

- **Question.** How large are the tax benefits of corporate debt, and do firms use enough debt to exploit those benefits?
- **Core contribution.** Instead of valuing debt tax shields as a statutory tax rate times the amount of debt, Graham builds firm-specific tax benefit functions and integrates under them.
- **Key object.** A firm's tax benefit function gives the marginal tax benefit of an additional dollar of interest deductions at different levels of interest expense.
- **Main measurement result.** The capitalized gross tax benefit of debt is about **9.7 percent of firm value** for the typical firm. Once the personal tax penalty on interest income is included, the net benefit falls to roughly **4–7 percent of firm value**.
- **Kink measure.** The paper defines *kink* as the ratio of the interest deduction level where the marginal benefit function first starts to decline to the firm's actual interest expense.
- **Debt conservatism.** A high-kink firm could increase interest deductions substantially before losing full marginal tax benefits. Graham interprets high kink as conservative debt policy.
- **Paradox.** Firms that appear able to use more debt safely—large, profitable, liquid firms with high Z-scores and low expected distress costs—often use debt conservatively.
- **Money left on the table.** If high-kink firms levered up to the kink, the typical firm could add about **15.7 percent of firm value** in gross tax benefits, or about **7.3 percent** after the personal tax penalty.
- **Persistence.** Debt conservatism is persistent: firms entering the most conservative kink quartile remain there at high rates over the next several years.
- **Economic takeaway.** Taxes create a large benefit to debt, but observed capital structures appear more conservative than a simple trade-off theory would predict. The paper therefore pushes the literature to explain why many firms leave large tax benefits unused.

2 Data and measurement

2.1 Sample and period

- The main sample is a COMPUSTAT firm-year panel from **1980 to 1994**.
- The full sample contains **87,643 firm-year observations**.
- The paper uses accounting data on taxable income, interest expense, non-debt tax shields, tax-loss carryforwards, investment tax credits, debt, market value, profitability, liquidity, and firm characteristics.
- Firm characteristics are winsorized at the top and bottom one percent to reduce the effect of extreme observations.

Interpretation. The empirical object is not a single firm-level tax rate. It is the whole shape of the tax benefit function at multiple hypothetical levels of interest deductions.

2.2 Simulated marginal tax rates

For each firm-year, Graham simulates marginal tax rates by forecasting taxable income and asking how much additional tax liability would arise from one extra dollar of income. A generic version is

$$\text{MTR}_{it} = \text{PV} [\text{Tax}_{it}(\text{income} + 1) - \text{Tax}_{it}(\text{income})]. \quad (1)$$

The calculation accounts for:

- tax-loss carrybacks;
- tax-loss carryforwards;
- investment tax credits;
- non-debt tax shields;
- the alternative minimum tax;
- uncertainty about future taxable income;
- state corporate taxes;
- personal taxes on interest income relative to equity income in the net-benefit calculations.

Interpretation. A firm can have a low current taxable income but still a positive marginal tax rate if losses can be carried back or forward. Conversely, additional interest can reduce future taxability and make marginal tax benefits decline.

2.3 Tax benefit function

Let x denote the level of interest deductions, expressed as a multiple of actual interest expense. Let $m_i(x)$ be the simulated marginal tax benefit for firm i at that level. The value of interest deductibility up to actual interest expense is approximately

$$TB_i = \int_0^1 m_i(x) I_i dx, \quad (2)$$

where I_i is actual interest expense.

The capitalized value is reported as a percentage of firm value:

$$\frac{PV(TB_i)}{V_i}. \quad (3)$$

Interpretation. The traditional formula $\tau_c D$ assumes a flat tax benefit function. Graham's approach allows the marginal benefit to fall once additional interest deductions reduce the probability of being taxable.

2.4 Gross and net tax benefits

- **Gross benefits** count the reduction in corporate and state tax liabilities caused by interest deductibility.
- **Net benefits** subtract the personal tax penalty: interest income is taxed more heavily than equity income for investors.
- The net calculation is a lower-bound style adjustment because it uses Treasury bill yields rather than an after-personal-tax discount rate.

Interpretation. The personal-tax adjustment matters. The gross tax benefit is large, but the net tax benefit is materially smaller.

2.5 Kink and ZeroBen

The paper defines

$$\text{Kink}_i = \frac{I_i^{\text{kink}}}{I_i}, \quad (4)$$

where I_i^{kink} is the amount of interest at which the marginal tax benefit first begins to decline.

It also defines

$$\text{ZeroBen}_i = \frac{I_i^0}{I_i}, \quad (5)$$

where I_i^0 is the amount of interest at which the marginal benefit of debt equals zero.

- If $\text{Kink}_i < 1$, the firm is already on the downward-sloping part of its benefit function. It is using debt aggressively from a tax perspective.
- If $\text{Kink}_i > 1$, the firm could add interest deductions and still receive full marginal tax benefits. It is using debt conservatively.
- If ZeroBen_i is high, the firm could add even more interest before marginal tax benefits fall to zero.

Interpretation. Kink is the paper’s central empirical measure of debt conservatism. It uses the *shape* of the benefit function, not just the amount of debt.

2.6 Firm characteristics used to explain kink

The paper relates kink to variables intended to proxy for costs and constraints of debt:

- distress risk: modified Altman Z-score, earnings volatility, negative owners’ equity, NOLs;
- information and payout policy: dividend dummy, no-dividend dummy;
- growth options and collateral: q-ratio, R&D-to-sales, PPE-to-assets;
- liquidity and size: current ratio, quick ratio, real sales, market value;
- product-market considerations: industry dummies, Herfindahl concentration;
- future financing needs: acquisitions and capital expenditures;
- managerial entrenchment variables from governance data.

Interpretation. If a standard trade-off theory fully explains leverage, high-kink firms should have high expected costs of debt. Graham finds that many high-kink firms instead look financially safe.

3 Models and empirical specifications

3.1 Benefit-function valuation

The paper values the tax benefit of debt by integrating under the marginal benefit curve:

$$\text{Tax Benefit}_{it} = \int_0^{I_{it}} MB_{it}(s) ds. \quad (6)$$

For reporting as a firm-value percentage:

$$\text{Tax Benefit Ratio}_{it} = \frac{\text{PV} \left(\int_0^{I_{it}} MB_{it}(s) ds \right)}{V_{it}}. \quad (7)$$

Interpretation. The method gives the area under the curve rather than assuming every dollar of interest gets the same statutory tax benefit.

3.2 Money left on the table

For firms with $\text{Kink}_i > 1$, Graham estimates the additional tax benefits from leveraging to the kink:

$$\text{Forgone Benefit}_{it} = \int_{I_{it}}^{I_{it}^{\text{kink}}} MB_{it}(s) ds. \quad (8)$$

Interpretation. This is a lower-bound-style estimate of potential value if the cost of debt does not rise before the kink. In the trade-off framework, the marginal cost curve should not cross the benefit curve on the flat portion.

3.3 Debt conservatism regression

The main cross-sectional regression is a Tobit model because kink is bounded in the construction:

$$\text{Kink}_{it}^* = X'_{it}\beta + u_{it}, \quad (9)$$

with observed kink censored into the reported support. The vector X_{it} includes profitability, distress proxies, liquidity, size, collateral, growth options, industry dummies, future investment variables, and governance variables.

Interpretation. The regression asks whether firms with high debt costs have higher kink. A positive coefficient means the variable is associated with more debt conservatism.

3.4 Persistence of debt conservatism

The persistence test follows firms after they first enter the highest kink quartile. Let Q_{it} be the kink quartile, where quartile 4 is the most conservative. The table reports

$$P(Q_{i,t+h} = q \mid Q_{it} = 4 \text{ for first time}). \quad (10)$$

Interpretation. Persistence matters because a high kink could be a temporary artifact of a short-lived earnings shock. The paper finds debt conservatism lasts long enough to look like a policy pattern.

3.5 Appendix marginal-tax-rate simulation

The appendix illustrates marginal-tax-rate simulation using a hypothetical income stream. For one extra dollar of current income, the marginal tax rate equals the present value of incremental tax liabilities:

$$\text{MTR}_t = \sum_{j \geq 0} \frac{\Delta \text{Tax}_{t+j}}{(1+r)^j}. \quad (11)$$

Interpretation. This is why marginal tax rates can differ from statutory rates: a current dollar can affect carrybacks, carryforwards, and future tax liabilities.

4 Identification and empirical design

4.1 What is identified

- The paper primarily identifies the *magnitude* of tax benefits and the *relative conservatism* of debt usage.
- It does not use an exogenous treatment shock to estimate a causal effect of taxes on debt issuance.
- The central empirical strategy is measurement: build firm-specific tax benefit functions, compute kink, and compare kink to firm characteristics.

Interpretation. This is a measurement-and-inference paper rather than a natural-experiment paper. Its strength is the careful construction of tax rates and benefit functions.

4.2 Why simulated tax rates are useful

- Statutory tax rates are not the same as marginal tax rates.
- Loss carrybacks and carryforwards make tax status dynamic.
- Non-debt tax shields and credits can crowd out interest deductions.
- Simulating future taxable income gives a more realistic expected marginal tax benefit.

Interpretation. A firm with current losses may still value interest deductions, while a profitable firm may lose value from extra deductions if they push it toward loss carryforwards.

4.3 How the kink measure speaks to optimal capital structure

- In a trade-off theory, firms borrow until marginal tax benefits are balanced by marginal costs of debt.
- If a firm is far to the left of the kink, its marginal tax benefits are still flat and high.
- Unless marginal debt costs are already rising sharply, this implies that the firm could increase debt and create tax value.

Interpretation. Kink does not directly measure the cost of debt. It measures whether the tax-benefit side has started to decline. High kink therefore raises the question: what cost or constraint prevents more debt?

4.4 Main empirical tension

- Some high-kink firms have growth options, intangible assets, or sensitive product-market positions, which are plausible reasons to preserve debt capacity.
- But many high-kink firms are large, profitable, liquid, dividend-paying, and have high Z-scores.
- These firms look like they face low ex ante distress costs, yet they use debt conservatively.

Interpretation. The paper does not deny that costs of debt matter. It argues that observed conservatism is too large to be explained only by standard distress-cost proxies.

4.5 What the paper cannot fully prove

- It cannot observe the full marginal cost function of debt.
- The personal-tax adjustment requires assumptions about implicit personal tax rates.
- State taxes are measured with principal-location information even though firms operate across states.
- The kink measure is capped, which compresses the right tail of debt conservatism.
- The analysis does not directly observe managerial objectives or private costs of financial distress.

Interpretation. The evidence shows that tax benefits are large and that many firms appear conservative relative to those benefits. It does not completely identify why those firms avoid debt.

5 Tables and figures

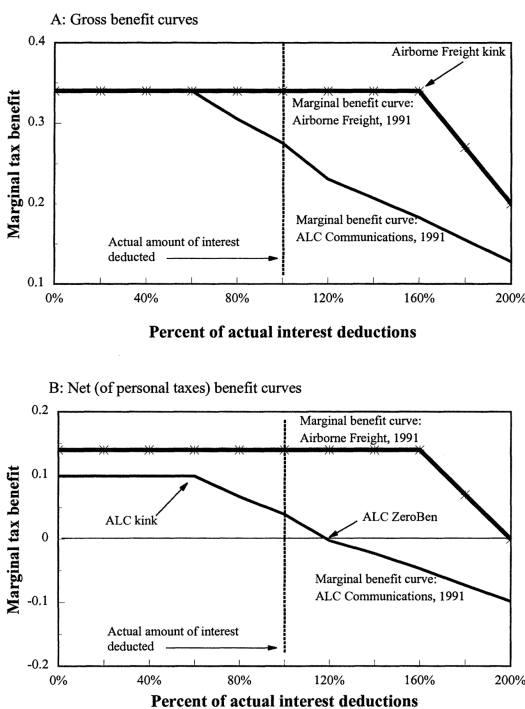
The original figures and tables are directly cropped from the paper. Adjacent visual material is grouped to save space and improve comparison.

5.1 Figure 1 and Table II: marginal benefit functions, kink, and debt conservatism

Read:

- Figure 1 shows marginal benefit curves for Airborne Freight and ALC Communications.
- Airborne Freight has a flat marginal benefit curve over a wide range and a kink around 1.6: it could raise interest deductions to about 160 percent of actual interest before marginal benefits decline.
- ALC Communications is already on a declining marginal-benefit curve and has a much lower kink.
- In the net-of-personal-tax panel, ALC's marginal benefit eventually becomes negative; this is the ZeroBen concept.
- Table II maps kink groups into standardized kink and observation counts. Many firms have high kink values; the 8.0 group alone has 11,519 observations.

Economic message. Kink turns the shape of a firm's tax benefit function into a measure of debt conservatism. High-kink firms are far from exhausting the tax benefits of interest deductions.



(a) Figure 1: marginal benefit curves.

Table II
Relation between the Kink in a Firm's Tax Benefit Function and Standardized Kink

The firm-year observations are grouped according to where a firm's marginal benefit function of interest deductions becomes downward sloping (i.e., the "kink" in the benefit function, where the benefit of incremental interest deductions begins to decline). The benefit function for a firm in the "kink = 1.6" group is not downward sloping for interest deductions 1.6 times those actually taken but becomes downward sloping for interest deductions greater than 1.6 times those actually taken. A firm in the group "kink = 0.2" has a downward-sloping marginal benefit curve for interest deductions greater than 20 percent of those actually taken. Firms in the 0.0* and 0.0 groups have downward-sloping benefit curves for the first dollar of interest expense. Firms in 0.0* (0.0) have a gross benefit less than (greater than or equal to) \$0.15 for the first dollar of interest. *Standardized kink* is dollars of interest associated with the kink, divided by the standard deviation of earnings.

Kink	Standardized Kink	Number of Obs.
0.0*	0.00	16,983
0.0	0.00	10,855
0.2	0.14	2,763
0.4	0.31	2,241
0.6	0.52	2,129
0.8	0.83	2,202
1.0	1.36	2,539
1.2	1.64	5,052
1.6	2.11	4,378
2.0	1.75	9,496
3.0	2.17	6,580
4.0	1.91	4,376
5.0	1.77	2,810
6.0	2.09	2,074
7.0	1.95	1,646
8.0	1.10	11,519

(b) Table II: kink and standardized kink.

5.2 Table I: summary statistics

Read:

- The full sample contains 87,643 firm-year observations from 1980 to 1994.

- The mean simulated before-financing tax rate net of personal taxes is 0.042; the median is 0.094.
- The mean kink is 2.356 and the median is 1.200.
- The mean ZeroBen is 3.540 and the median is 3.000.
- Firms vary substantially in distress risk, liquidity, growth options, and non-debt tax shields.

Economic message. The typical firm appears able to more than double interest deductions before marginal tax benefits begin to fall, suggesting substantial debt conservatism.

Table I
Summary Statistics

$MTR_{it}^{0\%}$ is the simulated before-financing tax rate, based on earnings before interest and taxes, net of the personal tax penalty. $MTR_{it}^{100\%}$ is the simulated marginal tax rate based on earnings before taxes, net of the personal tax penalty. $Z\text{-score}$ is the modified Altman's (1968) Z -score. $ECOST$ is the standard deviation of the first difference in taxable earnings divided by assets, the quotient times the sum of advertising, research, and development expenses divided by sales. $NODIV$ is a dummy variable equal to one if the firm does not pay dividends. $OENEG$ is a dummy variable equal to one if the firm has negative owners' equity. The $q\text{-ratio}$ is preferred stock plus market value of common equity plus net short-term liabilities, the sum divided by total assets. $PPE\text{-to-assets}$ is net property, plant, and equipment divided by total assets. $Ln(\text{real sales})$ is the natural log of real sales (where sales are deflated by the implicit price deflator), with sales expressed in millions of dollars. The $current\ ratio$ is short-term assets divided by short-term liabilities. $Return\text{-on-assets}$ is operating cash flow divided by total assets. RDS is the research and development expense to sales ratio, set equal to zero if the numerator is missing. $NOL\ dummy$ is equal to one if firm has book NOL carryforwards and is zero otherwise. $CYCLICAL$ is the standard deviation of operating earnings divided by mean assets first calculated for each firm, then averaged across firms within two-digit SIC codes. $Acquisitions$ and $capital\ expenditures$ are the two-year sums of acquisitions or capital expenditures, respectively, deflated by total assets. $Asset\ Herfindahl$ is an industry-wide Herfindahl index based on assets. $Kink$ is the amount of interest at the point where the marginal benefit function becomes downward sloping, as a proportion of actual interest expense. $ZeroBen$ is the amount of interest at the point where the marginal tax benefit of debt is zero, as a proportion of actual interest expense. The full sample has 87,643 observations from 1980 to 1994.

	Mean	Std. Dev.	Median	Minimum	Maximum
$MTR_{it}^{0\%}$	0.042	0.154	0.094	-0.427	0.302
$MTR_{it}^{100\%}$	0.006	0.177	0.075	-0.427	0.302
$Z\text{-score}$	1.011	4.186	1.889	-25.76	6.204
$ECOST$	0.994	4.463	0.010	0.000	36.46
$NODIV$	0.569	0.495	1.000	0.000	1.000
$OENEG$	0.074	0.262	0.000	0.000	1.000
$q\text{-ratio}$	1.034	1.693	0.513	-0.332	11.01
$PPE\text{-to-assets}$	0.330	0.262	0.270	0.000	0.930
$Ln(\text{real sales})$	-0.535	2.586	-0.390	-7.569	4.944
$Current\ ratio$	2.490	2.975	1.755	0.000	21.58
$Return\text{-on-assets}$	0.055	0.237	0.102	-1.343	0.413
RDS	0.041	0.150	0.000	0.000	1.000
$NOL\ dummy$	0.273	0.445	0.000	0.000	1.000
$CYCLICAL$	0.178	0.186	0.149	0.013	1.825
$Acquisitions$	0.023	0.082	0.000	-0.494	3.207
$Capital\ expenditures$	0.076	0.199	0.049	-0.500	31.00
$Asset\ Herfindahl$	0.271	0.284	0.155	0.015	1.000
$Kink$	2.356	3.111	1.200	0.000	8.000
$Zero\ benefit$	3.540	3.311	3.000	0.000	8.000

Table I: summary statistics.

5.3 Figure 2 and Table III: aggregate tax benefits of debt

Read:

- Table III shows that aggregate gross tax benefits rise from about \$50.8 billion in 1980 to \$94.8 billion in 1994.
- Gross tax benefits average about 9.7 percent of firm value in the paper's discussion.
- Net tax benefits are smaller, around 3–5 percent of firm value across many years after accounting for the personal tax penalty.
- Figure 2 shows that the gross tax benefit is fairly stable around 10 percent of firm value, while net benefits are lower.

- The “gross money left on the table” line declines sharply over time but remains positive.

Economic message. Interest deductibility is economically large. Even after personal taxes, the tax benefit of debt is not a second-order issue.

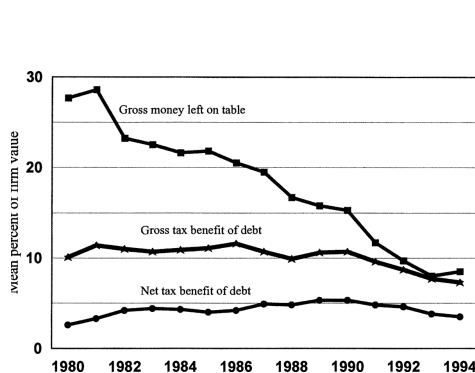


Figure 2: tax benefits of debt.

Table III
The Aggregate Tax Benefit of Debt

Gross benefits equal the area under each firm's gross benefit curve (up to the point of actual interest expense), aggregated across firms. Gross benefits measure the reduction in corporate and state tax liabilities occurring because interest expense is tax deductible. Net benefits equal gross benefits minus the personal tax penalty. That is, net benefits are reduced to account for the fact that firms must pay a higher risk-adjusted return on debt than on equity, because investors demand a higher return to compensate for the higher rate of personal taxation on interest income, relative to equity income. The *Total* and *Per Firm* columns express the annual tax benefits of debt. The *Percent of Firm Value* columns express the capitalized tax benefit of debt aggregated across firms, expressed as a percentage of aggregate firm value. *Zero Benefit* is the amount of interest for which the marginal tax benefit of debt equals zero, expressed as a proportion of actual interest expense. *Kink* is the amount of interest where the marginal benefit function becomes downward sloping, expressed as a proportion of actual interest expense.

	Gross Benefit			Net Benefit			Zero Benefit	Kink	N
	Total \$ (millions) annual	Per Firm \$ (millions) annual	Percent of Firm Value Capitalized	Total \$ (millions) annual	Per Firm \$ (millions) annual	Percent of Firm Value Capitalized			
1980	50,880	9.5	10.1	13,156	2.5	2.6	3.79	3.10	5,335
1981	63,196	12.1	11.4	18,114	3.5	3.3	3.68	2.98	5,215
1982	65,402	12.4	11.0	24,848	4.7	4.2	3.73	2.69	5,254
1983	62,447	11.8	10.7	25,528	4.8	4.4	3.74	2.68	5,281
1984	78,736	12.6	10.9	27,111	5.0	4.3	3.80	2.75	5,461
1985	75,013	13.6	11.1	27,310	4.9	4.0	3.55	2.51	5,524
1986	83,869	15.1	11.6	30,382	5.5	4.2	3.27	2.39	5,564
1987	81,106	13.6	10.7	36,980	6.2	4.9	3.53	2.35	5,971
1988	89,593	14.7	9.9	43,559	7.1	4.8	3.70	2.30	6,115
1989	109,292	17.9	10.6	54,351	8.9	5.3	3.65	2.24	6,108
1990	113,605	18.7	10.7	56,480	9.3	5.3	3.50	2.08	6,087
1991	107,411	17.5	9.6	53,661	8.8	4.8	3.42	1.99	6,123
1992	98,345	15.7	8.7	51,419	8.2	4.6	3.53	2.07	6,282
1993	84,937	13.1	7.7	42,168	6.6	3.8	3.11	1.71	6,479
1994	94,770	13.8	7.3	45,939	6.7	3.5	3.27	1.94	6,849

Table III: aggregate tax benefit of debt.

5.4 Table IV: individual firms and leveraged buyouts

Read:

- Boeing and Intel illustrate firms that are often profitable and can sustain high kink values.
- Pacific Gas & Electric, a regulated utility, has stable tax benefits and low kink around 2.0.
- RJR Nabisco and Safeway show what happens around leveraged buyouts: leverage rises sharply and kink falls.
- After the LBOs, tax benefits become much larger as a percentage of assets.
- The case evidence supports the idea that firms can materially increase tax benefits by leveraging up.

Economic message. The individual examples make the abstract tax-benefit functions concrete. Conservative firms can remain far from the kink, while LBO firms move closer to the high-debt region and realize larger tax benefits.

Table IV
The Tax Benefit of Debt and Other Characteristics for Some Individual Firms

Kink is the amount of interest required to make a firm's marginal benefit function become downward sloping expressed as a proportion of actual interest expense. *Gross benefit/market value* is the present value of gross tax savings expressed as a percentage of market value (where market value is book assets minus book equity plus market equity). *Net benefit/market value* is the present value of tax savings minus the personal tax penalty expressed as a percentage of market value. *Net benefit/total assets* is the present value of tax savings minus the personal tax penalty expressed as a percentage of book assets. *Debt/assets* is the book value of debt divided by the book value of assets. *Credit rating* is the firm's credit rating on senior debt as assigned by Standard and Poor. NA means not available. *EBIT* is earnings before interest and taxes. *MTR^{100%}* is the simulated marginal tax rate based on earnings before taxes, net of the personal tax penalty. Boeing is the representative "always profitable" firm. Intel is a typical "usually profitable" firm. Pacific Gas & Electric is a regulated electric utility. RJR Nabisco and Safeway both completed LBOs during the sample period; an asterisk * indicates the year they completed their respective leveraged buyouts.

Company	Characteristic	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993
Boeing	Kink	7.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	4.0	4.0	7.0
	Gross benefit/market value (%)	2.0	1.6	0.8	1.1	1.0	0.6	0.5	0.5	1.1	2.6	3.8
	Net benefit/market value (%)	0.7	0.5	0.2	0.3	0.4	0.3	0.3	0.2	0.4	1.0	1.5
	Debt/assets (%)	5.1	3.5	0.4	2.5	2.1	2.0	2.1	2.2	8.3	9.9	12.9
	Credit rating	NA	NA	AA-	AA-	AA-	AA-	AA-	AA	AA	AA	AA
	EBIT (millions)	517	605	883	1055	685	846	1404	2000	2261	848	2010
	MTR ^{100%} (%)	14.7	13.1	13.0	13.0	16.2	15.7	16.5	15.2	14.0	13.2	13.4
Intel	Kink	5.0	8.0	0.0	2.0	1.0	5.0	5.0	6.0	7.0	8.0	8.0
	Gross benefit/market value (%)	1.4	1.8	0.0	2.7	0.0	5.7	5.7	4.9	3.4	1.5	1.0
	Net benefit/market value (%)	0.3	0.4	0.0	0.6	0.0	2.1	2.1	1.8	1.2	0.6	0.4
	Debt/assets (%)	12.4	30.4	16.7	19.2	28.9	19.6	14.3	11.6	8.5	6.9	8.1
	Credit rating	NA	NA	A	A	A	A	A	A	A	A	A+
	EBIT (millions)	202	323	31	-46	466	721	780	1089	1299	1698	3573
	MTR ^{100%} (%)	7.5	7.5	-9.3	7.5	8.2	9.9	9.9	9.9	9.9	10.6	108

Pacific Gas and Electric	Kink	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
	Gross benefit/market value (%)	19.0	18.4	19.3	18.9	18.6	15.9	15.8	14.5	12.6	13.2	13.4
	Net benefit/market value (%)	10.2	9.5	10.1	10.0	12.7	11.8	11.8	10.8	8.3	8.7	8.8
	Debt/assets (%)	46.3	45.0	43.5	42.7	41.8	40.2	40.0	40.7	41.1	40.8	37.8
	Credit rating	NA	NA	A+	A+	A+	A	A	A	A	A	A
	EBIT (millions)	1965	2257	2596	2807	2086	1875	2424	2761	2703	2855	3000
	MTR ^{100%} (%)	23.5	22.3	22.8	23.0	25.5	24.2	24.2	24.2	21.1	20.9	21.7
RJR Nabisco	Kink	6.0	7.0	6.0	6.0	5.0	5.0	2.0 ^a	1.2	1.2	1.2	1.2
	Gross benefit/market value (%)	8.4	8.1	10.7	15.2	11.6	7.8	23.8	NA	28.0	20.6	21.1
	Net benefit/total assets (%)	3.5	3.9	4.8	8.2	7.4	7.5	14.7	15.3	11.3	7.8	7.2
	Debt/assets (%)	16.2	17.1	33.6	34.0	26.7	32.3	68.0	56.7	44.6	43.8	40.0
	Credit rating	NA	NA	NA	NA	NA	BB	BB	BB	BBB-	BBB-	BBB-
	EBIT (millions)	1701	2448	2778	2532	2784	2949	2772	2990	2883	2379	1828
	MTR ^{100%} (%)	16.2	16.4	16.1	15.9	17.3	15.9	9.9	9.9	9.9	9.9	10.0
Safeway	Kink	2.0	2.0	2.0	1.0 ^a	0.8	1.0	1.0	1.0	1.0	1.0	1.2
	Gross benefit/market value (%)	18.2	15.6	18.3	17.2	NA	NA	28.8	24.8	22.8	20.9	20.9
	Net benefit/total assets (%)	6.6	6.7	7.5	12.2	18.1	13.5	13.0	12.8	11.5	9.8	9.8
	Debt/assets (%)	31.3	32.8	30.8	76.3	75.4	72.0	68.7	65.1	59.2	58.3	53.0
	Credit rating	NA	NA	A	B+	B+	B+	BB-	BB	BB	BB	BB
	EBIT (millions)	510	530	556	522	-98	520	527	634	638	467	595
	MTR ^{100%} (%)	18.0	17.7	17.2	19.5	2.6	9.9	9.9	9.9	9.9	9.9	9.9

5.5 Table V: relation between kink and firm characteristics

Read:

- Firms with low kink have high debt-to-value, low Z-scores, high no-dividend rates, and negative or low profitability.
- Firms with high kink have much lower debt-to-value, higher Z-scores, more liquidity, larger market value, and higher profitability.
- For example, the 8.0 kink group has debt-to-value of 0.115, Z-score of 3.058, current ratio of 3.827, and ROA of 0.182.
- The table therefore shows a paradox: high-kink firms look safer, not riskier.

Economic message. Many conservative debt users do not look constrained by standard distress-cost proxies. This is the central empirical puzzle of the paper.

Table V
Relation between the Kink in a Firm's Tax Benefit Function and Firm Characteristics

The firm-year observations are grouped according to where a firm's marginal benefit function of interest deductions becomes downward sloping (i.e., the "kink" in the benefit function, where the benefits of incremental interest begin to decline). The benefit function for a firm in the "kink = 1.6" group becomes downward sloping for interest deductions greater than 1.6 times those actually taken. Firms in the 0.0* and 0.0 groups have downward-sloping benefit curves for the first dollar of interest expense. Firms in 0.0* (0.0) have a gross benefit less than (greater than or equal to) \$0.15 for the first dollar of interest. The table presents mean values for firm characteristics for the years 1980 to 1994. *Debt-to-value* is long-term debt plus debt in current liabilities, the sum divided by the market value of the firm. The *market value* of the firm is book assets minus book equity plus market equity. *Z-score* is the modified Altman's (1968) Z-score. *NODIV* is a dummy variable equal to one if the firm does not pay dividends. *RDS* is research and development expense to sales ratio, set equal to zero if the numerator is missing. The *q-ratio* is preferred stock plus market value of common equity plus net short-term liabilities, the sum divided by total assets. *PPE-to-assets* is net property, plant, and equipment divided by total assets. *Ln(real sales)* is the natural log of real sales (where sales are deflated by the implicit price deflator), with sales expressed in millions of dollars. The *current ratio* is short-term assets divided by short-term liabilities. *CYC* is the standard deviation of operating earnings divided by mean assets first calculated for each firm, then averaged across firms within two-digit SIC codes. *ROA*, return-on-assets, is operating cash flow divided by total assets.

Kink	Debt-to-Value	Z-Score	NODIV	RDS	q-ratio	PPE-to-Assets	Ln(real Sales)	Market Value	Current Ratio	CYC	ROA
0.0*	0.328	-2.902	0.911	0.137	2.095	0.305	-2.901	328	2.934	0.247	-0.198
0.0	0.400	0.155	0.843	0.035	0.926	0.312	-1.722	590	2.383	0.216	0.020
0.2	0.441	1.215	0.816	0.020	0.585	0.322	-1.110	684	2.189	0.189	0.050
0.4	0.459	1.651	0.779	0.015	0.497	0.335	-0.769	673	1.990	0.182	0.071
0.6	0.458	1.761	0.752	0.013	0.408	0.334	-0.527	1005	1.989	0.176	0.082
0.8	0.471	1.941	0.715	0.012	0.384	0.339	-0.396	1042	1.876	0.167	0.088
1.0	0.518	2.001	0.621	0.010	0.418	0.398	-0.019	2352	1.778	0.164	0.095
1.2	0.498	2.131	0.541	0.009	0.380	0.311	0.178	1996	1.825	0.149	0.105
1.6	0.414	2.130	0.447	0.010	0.426	0.364	0.395	2131	1.849	0.148	0.114
2.0	0.368	2.155	0.335	0.010	0.500	0.422	0.763	2159	1.878	0.138	0.127
3.0	0.317	2.342	0.303	0.013	0.593	0.406	0.876	2024	2.055	0.141	0.143
4.0	0.257	2.599	0.284	0.016	0.766	0.382	0.971	2473	2.279	0.145	0.156
5.0	0.221	2.713	0.270	0.019	0.832	0.334	0.961	2707	2.464	0.146	0.159
6.0	0.192	2.741	0.279	0.019	0.901	0.318	0.933	3007	2.579	0.143	0.161
7.0	0.176	2.833	0.283	0.020	0.963	0.323	0.758	3235	2.623	0.145	0.171
8.0	0.115	3.058	0.315	0.024	1.480	0.254	0.169	2103	3.827	0.151	0.182

Table V: kink and firm characteristics.

5.6 Table VI: Tobit regressions using kink as the dependent variable

Read:

- The Tobit regressions confirm the univariate patterns in a multivariate setting.
- Dividend-paying, profitable, high-Z-score, and liquid firms are associated with higher kink, meaning more debt conservatism.
- Some variables consistent with debt-cost explanations also matter: firms with growth options, R&D, low collateral, and sensitive product-market positions can be conservative.
- Future acquisitions and capital expenditures have positive coefficients in some specifications, but the economic effect is small.
- Governance and entrenchment variables do not provide a complete explanation once full controls are included.

Economic message. The regression evidence is mixed. Some conservative debt behavior is explainable by growth options and intangible assets, but much of it remains puzzling because safe and profitable firms are also conservative.

Tobit Regressions Using the Kink in the Benefit Function as Dependent Variable

$I(.)$ defines a dummy variable equal to one if a firm is in a certain industry or meets the listed condition. *CYCLICAL* is the standard deviation of operating earnings divided by mean assets first calculated for each firm, then averaged across firms within two-digit SIC codes. *Return-on assets* is operating cash flow divided by total assets. *Ln(real sales)* is the natural log of real sales (where sales are deflated by the implicit price deflator), with sales expressed in millions of dollars. *Z-score* is the modified Altman's (1968) *Z-score*. *ECOST* is the standard deviation of the first difference in taxable earnings divided by assets, the quotient times the sum of advertising, research, and development expenses divided by sales. The *current ratio* is short-term assets divided by short-term liabilities. The *quick ratio* is cash, short-term investments, and receivables, the sum divided by current liabilities. *PPE-to-assets* is net property, plant, and equipment divided by total assets. The *q-ratio* is preferred stock plus market value of common equity plus net short-term liabilities, the sum divided by total assets. *Advertising-to-sales* is the ratio of advertising expense to sales and *R&D-to-sales* is research and development expense to sales ratio, with either variable set equal to zero if its numerator is missing. The *Herfindahl* indices are calculated within two-digit SIC codes for *assets* and *sales*, respectively. *Capital expenditures and acquisition* are from the statement of cash flows and measure the sum of investing activities in Years $t + 1$ and $t + 2$. *CEOSTOCK* is the percentage of common shares owned directly by the CEO. *BDSTOCK* is the percentage of common shares held by non-CEO board members. *CEOOPT* measures vested options held by the CEO expressed as a percentage of total common shares. *YRSCFO* is the log of the number of years the CEO has been chief executive. *BOARDSZ* is the log of the number of directors, and *PCTOUT* measures the percentage of directors that are outsiders.

	Estimated Coefficients							
	Column VIa		Column VIb		Column VIc		Column VI d	
	Variable	(Variable) ²	Variable	(Variable) ²	Variable		Variable	(Variable) ²
Intercept	0.354*		0.412*		1.635*		1.260*	
I(No dividend)	-0.483*		-0.384*				-0.273*	
I(Negative owners' equity)	-0.426*		-0.456*				-0.076	
I(NOL carryforward)	-0.459*		-0.456*				-0.277*	
I(chemicals and allied products)	0.067*		0.054**				0.069	
I(computers)	0.276*		0.263*				0.426*	
I(semiconductors)	0.040		0.008				0.318**	
I(wholesale chemicals)	0.665*						9.835	
I(aircraft/guided space vehicles)	0.052		0.056				0.285*	
I(other sensitive industries)	0.114*		0.117*				0.194*	
CYCLICAL	-0.308*		-0.176*				-0.992*	
Return on assets	2.234*	2.700*	2.130*	2.887*	7.029*		1.603*	4.413*
Ln(real sales)	0.048*	-0.002**	0.049*	-0.002*	-0.092*		-0.330*	0.046
Z-score	0.126*	0.004*	0.124*	0.004*			0.513*	-0.063*
ECOST	-0.117*	0.003*	-0.119*	0.003*			-0.647**	-0.324***
Current ratio	0.127*	-0.009*	0.135*	-0.009*			-0.009	0.002
Quick ratio	0.253*	-0.006*	0.250*	-0.006*			0.262*	-0.014
PPE-to-assets	-1.395*	1.375*	-1.428	1.355*	-0.865*		-2.194*	2.276*
Q-ratio	0.503*	-0.046*	0.527*	-0.047*			0.573*	-0.579*
Advertising-to-sales	-0.155	11.74*	-0.558	14.39*			-0.479	7.071
R&D-to-sales	1.466*	-0.891*	1.417*	-1.034*	0.899**		3.285*	-6.581*
Asset Herfindahl			-0.515*	0.569*				
Sales Herfindahl			0.247	-0.323***				
Capital expenditures			0.215***	-0.078				
Acquisitions			0.749*	-2.176*				
CEOSTOCK					-0.591*		0.071	
CEOOPT					-11.46**		-6.705	
YRSCFO					0.039*		-0.003	
BOARDSZ					-0.105**		0.014	
PCTOUT					-0.294*		-0.351*	
BDSTOCK					0.213		0.014	
OLS R ²	50.2%		49.5%		44.5%		61.2%	
Number of observations.	65,373		55,644		2,910		2,773	

*, **, *** Significant at the 1 percent, 5 percent, and 10 percent level, respectively.

Table VI: Tobit regressions using kink as the dependent variable.

5.7 Table VII: persistence of debt conservatism

Read:

- Firms are followed after first entering the highest kink quartile.
- In Year 1, 82.4 percent remain in the most conservative quartile.
- In Year 2, 69.2 percent remain there.
- Even after five years, 39.2 percent remain in quartile 4, while only 13.8 percent have fallen to quartile 1.

Economic message. Debt conservatism is not just a one-year artifact. It is persistent enough to look like a stable policy or organizational characteristic.

Table VII
How Persistent Is Debt Conservatism?

This table shows the persistence of debt conservatism as defined by *kink*, where kink is the amount of interest required to make a firm's marginal benefit function become downward sloping expressed as a proportion of actual interest expense. A large value of kink indicates that a firm uses debt conservatively. The table follows firms after they first enter the highest (most conservative) quartile of kink. These high-quartile firms are followed over the subsequent five years. Year 0 is the year the firm enters quartile 4 (the most conservative). The number of firm-year observations is in parentheses.

	% in quartile 4	% in quartile 3	% in quartile 2	% in quartile 1
Year 0 (<i>N</i> = 4,960)	100.0			
Year 1 (<i>N</i> = 4,540)	82.4	10.3	4.9	2.5
Year 2 (<i>N</i> = 4,135)	69.2	16.2	9.4	5.2
Year 3 (<i>N</i> = 3,666)	56.0	21.6	13.6	8.7
Year 4 (<i>N</i> = 3,189)	47.0	25.1	16.8	11.1
Year 5 (<i>N</i> = 2,745)	39.2	26.6	20.4	13.8

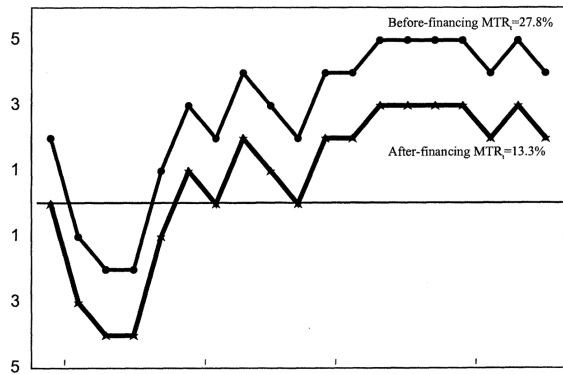
Table VII: persistence of debt conservatism.

5.8 Appendix Figure A1 and Table A1: simulated marginal tax rates

Read:

- Figure A1 shows a hypothetical income stream used to compute simulated marginal tax rates.
- The before-financing marginal tax rate in the example is 27.8 percent.
- The after-financing marginal tax rate falls to 13.3 percent because interest deductions reduce taxable income and change the use of loss carrybacks and carryforwards.
- Table A1 shows the tax liabilities in the base case, the extra-income case, and the incremental tax-liability case.

Economic message. Marginal tax rates are dynamic. The value of a deduction depends on whether it can be used now, carried back, carried forward, or crowded out by other tax shields.



(a) Figure A1: simulated marginal tax rate example.

Table A1

Calculating a marginal tax rate. The top panel shows taxes paid on the base case income stream, assuming a tax rate of 35 percent. (The base case income stream represents the top line in Figure A1.) The middle panel shows tax liabilities for the same firm if it earns one extra dollar in $t = 0$. The bottom panel shows the incremental tax liability associated with the extra $t = 0$ dollar. The marginal tax rate is the present value tax liability associated with the extra dollar of $t = 0$ income. The present value calculation at the bottom assumes a 10 percent discount rate. All numbers are dollars. The marginal tax rate is defined as the present value of incremental tax liabilities due from earning an extra dollar of $t = 0$ income and equals 27.8 percent ($0.278 = 0.35 - (0.35/1.1^2) + (0.35/1.1^5)$) in this table.

	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
Panel A: Base Case (top line in Figure A1)						
Income	2	-1	-2	-2	1	3
Tax-loss carryforward	0	0	1	3	2	0
Tax liability	0.70	-0.35	-0.35	0	0	0.35
Panel B: Earning an Extra Dollar in $t = 0$						
Income	3	-1	-2	-2	1	3
Tax-loss carryforward	0	0	0	2	1	0
Tax liability	1.05	-0.35	-0.70	0	0	0.70
Panel C: Incremental Tax Liabilities from Earning an Extra Dollar in $t = 0$						
Tax liability	0.35	0	-0.35	0	0	0.35

(b) Table A1: calculating the marginal tax rate.

6 Short summary

- Graham builds firm-specific marginal tax benefit functions for interest deductions.

- The value of debt tax shields is the area under the marginal benefit function, not simply statutory tax rate times debt.
- The gross capitalized tax benefit of debt is about 9.7 percent of firm value for the typical firm.
- Net of personal taxes, the benefit falls to roughly 4–7 percent of firm value.
- Kink measures the amount of interest a firm could deduct before marginal tax benefits begin to decline, relative to actual interest expense.
- The mean kink is 2.356 and the median kink is 1.2, indicating substantial unused tax capacity for many firms.
- Nearly half the sample could double interest deductions before ending up on the downward-sloping part of the benefit function.
- Conservative debt users are often large, profitable, liquid, and financially healthy.
- Growth options, intangible assets, product-market sensitivity, and future financing needs explain part of the conservatism.
- However, the overall pattern remains puzzling for a simple static trade-off theory.
- Levering high-kink firms to the kink could add about 15.7 percent of firm value in gross tax benefits and about 7.3 percent net of personal taxes.
- Debt conservatism is persistent across years.

7 Conclusion

7.1 Core conclusion

Graham shows that the tax benefits of debt are large and measurable once firm-specific tax benefit functions are constructed. Interest deductibility adds meaningful value, but many firms appear to use debt conservatively relative to their available tax benefits.

7.2 Economic intuition

The intuition is that interest deductions are valuable only when firms can use them. For low levels of debt, many firms receive the full marginal tax benefit. As interest deductions grow, expected taxable income falls and the marginal benefit declines. The kink reveals how far a firm can increase interest before reaching that declining region. A firm with a high kink is leaving tax capacity unused unless some offsetting cost or constraint explains its conservative policy.

7.3 Takeaway

The paper is foundational because it changes how researchers measure the tax benefit of debt. It also creates a central capital-structure puzzle: many financially healthy firms appear to leave large debt tax benefits on the table. Explaining that conservatism requires looking beyond simple tax-versus-bankruptcy trade-offs to product-market risk, growth options, financial flexibility, agency problems, and managerial preferences.